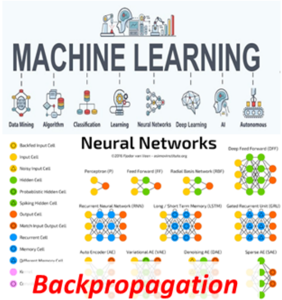
MH1100 Calculus I

Handout 8 - Derivatives III & IV

Synopsis

- The Chain Rule
- Implicit Differentiation
- Rates of Change
- Related Rates
- Linear Approximations
- Differentials

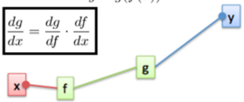
Application



Backpropagation

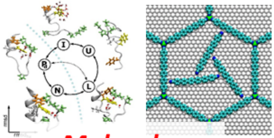
$$y = g(f(x))$$

$$\boxed{\frac{dg}{dx} = \frac{dg}{df} \cdot \frac{df}{dx}}$$

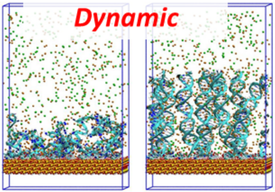


- ❖ Chain rules
- ❖ Implicit Differentiation

Various applications in natural and social sciences

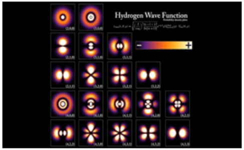


Molecular Dynamic



- ❖ Rates of Change
- ❖ Related Rates

Further readings:
Taylor series;
Fourier series;
Fast Fourier transform;
Spectral method ......



$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \ldots$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \ldots$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \ldots \quad for\,|x| < 1$$

$$\tan^{-1}(x) = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots \quad for\,|x| < 1$$

- ❖ Linear Approximations
- ❖ Differentials

The Chain Rule

The Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and F' is given by the product

$$F'(x) = f'(g(x)) \cdot g'(x)$$

In Leibniz notation, if $y = f(u)$ and $u = g(x)$, then

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}.$$

To use the Chain Rule:

- First, find and differentiate the **outside** function.
- Then, evaluate what you get at the point $g(x)$.
- Finally, multiply that by the derivative of the **inside** function, evaluated at x.

Example

Apply the chain rule to differentiate

$$h(x) = \sin\left(\frac{x^2 - 1}{x^2 + 1}\right).$$

Solution. This function is a composition $h(x) = f \circ g(x)$, with:

- The **outside** function: $f(u) = \sin u$,
- The **inside** function: $u = g(x) = \frac{x^2-1}{x^2+1}$.

Note that, by the differentiation rules we have developed, both of $f(u)$ and $g(x)$ are differentiable at every point of $\mathbb{R}$.

This means that we can use the Chain Rule to evaluate the derivative.

Solution

- **Step 1**: Differentiate the **outside** function $f(u) = \sin u$.

$$f'(u) = \cos u.$$

- **Step 2**: Evaluate the result of step 1 at $g(x)$ (the value of the **inside** function at the point x).

$$\cos\left(\frac{x^2-1}{x^2+1}\right)$$

- **Step 3**: Multiply the result of step 2 by the derivative of the inside function.

$$\begin{aligned}
\frac{dh}{dx} &= \cos\left(\frac{x^2-1}{x^2+1}\right)\cdot\frac{d}{dx}\left[\frac{x^2-1}{x^2+1}\right]\\
&= \cos\left(\frac{x^2-1}{x^2+1}\right)\cdot\frac{(2x)(x^2+1)-(2x)(x^2-1)}{(x^2+1)^2}\\
&= \cos\left(\frac{x^2-1}{x^2+1}\right)\cdot\frac{4x}{(x^2+1)^2}
\end{aligned}$$

The Chain Rule

Exercise

Differentiate (a) $y = \sin(x^2)$ and (b) $y = \sin^2 x$.

$y = \sin u$
$u = x^2$
$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx} = \cos u \cdot 2x = \cos(x^2)2x = 2x\cos x^2$

bring in expression for u

Solution.

(a) If $y = \sin(x^2)$, then outside function is sine function and inside function is square function. From Chain Rule, we have

$$\frac{dy}{dx} = \frac{d}{dx}\sin(x^2) = \cos(x^2) \cdot 2x = 2x\cos(x^2).$$

(b) Notice that $\sin^2 x = (\sin x)^2$. Here outside function is square function and inside function is sine function. From Chain Rule, we have

$$\frac{dy}{dx} = \frac{d}{dx}(\sin x)^2 = 2 \cdot (\sin x) \cdot \cos x = 2\sin x \cos x = \sin 2x.$$

The Chain Rule

If $y = [g(x)]^n$, we can write $y = f(u) = u^n$ where $u = g(x)$. By using Chain Rule and Power Rule, we get

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = nu^{n-1}\frac{du}{dx} = n\,[g(x)]^{n-1}\,g'(x)$$

The Power Rule Combined with the Chain Rule

If n is a real number and $u = g(x)$ is differentiable, then

$$\frac{d}{dx}(u^n) = nu^{n-1}\frac{du}{dx}$$

Alternatively,

$$\frac{d}{dx}[g(x)]^n = n\,[g(x)]^{n-1}\,g'(x)$$

The Chain Rule

Example

Differentiate $y = (x^3 - 1)^{100}$. u^{100} $u = x^3 - 1$

Solution. Taking $u = g(x) = x^3 - 1$ and $n = 100$, we have

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx}(x^3-1)^{100} = 100(x^3-1)^{99}\frac{d}{dx}(x^3-1)\\ &= 100(x^3-1)^{99}\cdot(3x^2) = 300x^2(x^3-1)^{99}.\end{aligned}$$

Example

Find $f'(x)$ if $f(x) = \frac{1}{\sqrt[3]{x^2+x+1}}$.

Solution. First, we can rewrite f: $f(x) = (x^2+x+1)^{-\frac{1}{3}}$.

Taking $u = g(x) = x^2+x+1$ and $n = -\frac{1}{3}$, we have

$$f'(x) = -\frac{1}{3}(x^2+x+1)^{-\frac{1}{3}-1}\frac{d}{dx}(x^2+x+1) = -\frac{1}{3}(x^2+x+1)^{-\frac{4}{3}}(2x+1).$$

The Chain Rule

$u(x) = g(h(x))$

$\rightarrow$ innermost layer

Suppose that $y = f(u)$, $u = g(w)$, and $w = h(x)$, where f, g, and h are differentiable functions.

$y = f[g(h(x))]$

To compute the derivative of y with respect to t, we can use Chain Rule twice:

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx} = \frac{dy}{du}\frac{du}{dw}\frac{dw}{dx}$$

$y = \cos(\tan(x))$

For example, if $f(x) = \sin(\cos(\tan x))$, then

$$\begin{aligned} f'(x) &= \cos(\cos(\tan x)) \cdot \frac{d}{dx}(\cos(\tan x)) \\ &= \cos(\cos(\tan x)) \cdot [-\sin(\tan x)]\frac{d}{dx}(\tan x) \\ &= -\cos(\cos(\tan x))\sin(\tan x)\sec^2 x. \end{aligned}$$

The Chain Rule

Example 8

Differentiate $y = \sqrt{\sec x^3}$. $[\sec(x^3)]^{\frac{1}{2}}$

Solution Here outsider function is square root function, middle function is secant function, and inside function is cubic function.

So we have

$$\begin{aligned}
\frac{dy}{dx} &= \frac{1}{2\sqrt{\sec x^3}} \cdot \frac{d}{dx}(\sec x^3) \\
&= \frac{1}{2\sqrt{\sec x^3}} \cdot \left[\sec x^3 \tan x^3\right] \cdot \frac{d}{dx}(x^3) \\
&= \frac{1}{2\sqrt{\sec x^3}} \cdot \left[\sec x^3 \tan x^3\right] \cdot (3x^2) \\
&= \frac{3x^2 \sec x^3 \tan x^3}{2\sqrt{\sec x^3}}
\end{aligned}$$

The Chain Rule

Exercise

Differentiate $y = \sin\left(\sqrt{\cos^2 x + 1}\right)$.

Solution. Using Chain Rule, we have

$$\begin{aligned}
\frac{dy}{dx} &= \cos\left(\sqrt{\cos^2 x + 1}\right) \cdot \frac{d}{dx}(\sqrt{\cos^2 x + 1}) \\
&= \cos\left(\sqrt{\cos^2 x + 1}\right) \cdot \frac{1}{2\sqrt{\cos^2 x + 1}} \cdot \frac{d}{dx}(\cos^2 x + 1) \\
&= \cos\left(\sqrt{\cos^2 x + 1}\right) \cdot \frac{1}{2\sqrt{\cos^2 x + 1}} \cdot (2\cos x) \cdot \frac{d}{dx}(\cos x) \\
&= \cos\left(\sqrt{\cos^2 x + 1}\right) \cdot \frac{1}{2\sqrt{\cos^2 x + 1}} \cdot (2\cos x) \cdot (-\sin x) \\
&= -\cos\left(\sqrt{\cos^2 x + 1}\right) \frac{\sin x \cos x}{\sqrt{\cos^2 x + 1}}
\end{aligned}$$

Part I. Property of differentiable functions

If $y = f(x)$ and x changes from a to $a + \Delta x$, we define the increment of y as

$$\Delta y = f(a + \Delta x) - f(a).$$

$\Delta x \to 0$
$\Delta y \to 0$

According to the definition of derivative, we have

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = f'(a).$$

So if we denote ϵ as

$$\epsilon = \frac{\Delta y}{\Delta x} - f'(a),$$

$\Delta x \to 0$, $\epsilon \to 0$, $\Delta y \to 0$

we will have

$$\Delta y = \underbrace{f'(a)\Delta x}_{\text{some major term}} + \epsilon \Delta x$$

and

$$\lim_{\Delta x \to 0} \epsilon = \lim_{\Delta x \to 0} \left(\frac{\Delta y}{\Delta x} - f'(a) \right) = f'(a) - f'(a) = 0.$$

Proof of The Chain Rule

Thus for a differentiable function f, we can have

$\Delta y \to 0$

$$\Delta y = f'(a)\Delta x + \epsilon \Delta x \qquad \text{where} \qquad \epsilon \to 0 \quad \text{as} \quad \Delta x \to 0.$$

This property of differentiable functions is what enables us to prove the Chain Rule.

Part II, Proof of the Chain Rule. $y = f[g(x)]$

Suppose $u = g(x)$ is differentiable at a and $y = f(u)$ is differentiable at $b = g(a)$. If Δx is an increment in x and Δu and Δy are the corresponding increments in u and y, then we can write

$$\Delta u = g'(a)\Delta x + \epsilon_1 \Delta x = \underline{\left[g'(a) + \epsilon_1\right] \Delta x} \quad \text{where} \quad \epsilon_1 \to 0 \quad \text{as} \quad \Delta x \to 0$$

$\Delta u \to 0$ also

and

$$\Delta y = f'(b)\Delta u + \epsilon_2 \Delta u = \left[f'(b) + \epsilon_2\right] \Delta u \quad \text{where} \quad \epsilon_2 \to 0 \quad \text{as} \quad \Delta u \to 0.$$

$\Delta y \to 0$ also

Proof of The Chain Rule

Substitute the expression for Δu into the expression for Δy, we get

$$\Delta y = \left[f'(b) + \epsilon_2\right] \cdot \left[g'(a) + \epsilon_1\right] \Delta x.$$

$\Delta x \to 0$
then $\varepsilon_1 \to 0$
then $\Delta u \to 0$
then $\varepsilon_2 \to 0$

So

calculate derivative

$$\frac{\Delta y}{\Delta x} = \left[f'(b) + \epsilon_2\right] \cdot \left[g'(a) + \epsilon_1\right].$$

As $\Delta x \to 0$, $\Delta u \to 0$. So both $\epsilon_1 \to 0$ and $\epsilon_2 \to 0$ as $\Delta x \to 0$.

Therefore

$$\begin{aligned}
\frac{dy}{dx} &= \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} \left[f'(b) + \epsilon_2\right] \cdot \left[g'(a) + \epsilon_1\right] \\
&= \lim_{\Delta x \to 0} \left[f'(b) + \epsilon_2^{\,\to 0}\right] \cdot \lim_{\Delta x \to 0} \left[g'(a) + \epsilon_1^{\,\to 0}\right] \\
&= f'(b)g'(a) = f'(g(a))g'(a).
\end{aligned}$$

$b = g(a)$ from previous slide

This proves the Chain Rule.

Implicit Differentiation

The functions that we have met so far can be described by expressing one variable explicitly in terms of another variable, for example

$$\text{(a)}\quad y = \sqrt{x^3+1} \qquad\qquad \text{(b)}\quad y = x\sin x$$

or, in general $y = f(x)$. Some functions, however, are defined implicitly by a relation between x and y such as

$$\text{(1)}\quad x^2 + y^2 = 25 \Rightarrow y = \pm\sqrt{25-x^2}$$

or

$$\text{(2)}\quad x^3 + y^3 = 6xy \Rightarrow y = f(x)$$

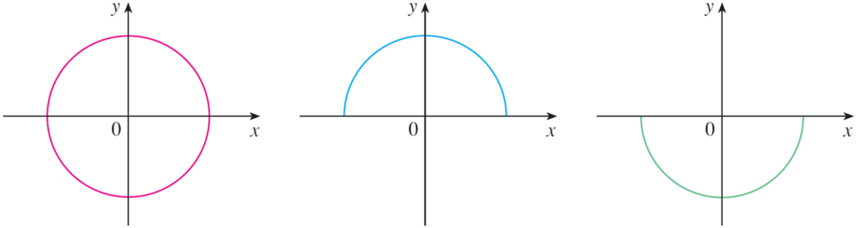
$\sin(x+y) + e^y = \tan(x^2+y^2) \rightarrow$ so complex, cannot represent in $y=f(x)$ form easily

Implicit Differentiation

In some cases, it is possible to solve such an equation for y as an explicit function (or several functions) of x.
For instance, if we solve Equation (1) for y, we get $y = \pm\sqrt{25 - x^2}$, and two of the functions determined by the implicit Equation (1) are

$$f(x) = \sqrt{25 - x^2} \quad \text{and} \quad g(x) = -\sqrt{25 - x^2}$$

The graphs of $f(x)$ and $g(x)$ are the upper and lower semicircles (of circle $x^2 + y^2 = 25$).



(a) $x^2 + y^2 = 25$ (b) $f(x) = \sqrt{25 - x^2}$ (c) $g(x) = -\sqrt{25 - x^2}$

Its not easy to solve Equation (2) for y (as a function of x) explicitly.

Fortunately, we don't need to solve an equation for y in terms of x in order to find the derivative of y. Instead we can use the method of **implicit differentiation**.

① Figure out what is the variable and function
② Do the differentiation on both sides of the eqn with respect to the variable.
③ Solve the eqn to find derivative expression

Implicit differentiation is to differentiate both sides of the equation with respect to x, and then solve the resulting equation for y'.

In the following examples and exercises, it is always assumed that y is a differentiable function of x. The method of implicit differentiation is applied to calculate y implicitly as a function of x.

Implicit Differentiation

Example

(a) If $x^2 + y^2 = 25$, find $\frac{dy}{dx}$. $\Rightarrow y(x)$

(b) Find an equation of the tangent to the circle $x^2 + y^2 = 25$ at the point (3, 4)

Solution

(a) Differentiate both sides of the equation

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0$$

Remembering that y is a function of x and using the Chain Rule, we have

$$\frac{d}{dx}(y^2) = \frac{d}{dy}(y^2)\frac{dy}{dx} = 2y\frac{dy}{dx}$$

Implicit Differentiation

Thus

$$2x + 2y\frac{dy}{dx} = 0$$

Now we solve this equation for $\frac{dy}{dx}$

$$\frac{dy}{dx} = -\frac{x}{y}$$

$\frac{dy}{dx} = f(x)$ usually
$\rightarrow$ but here have y
$\therefore$ implicit differentiation

(b) At the point (3, 4) we have $x = 3$ and $y = 4$, so

$$\frac{dy}{dx} = -\frac{3}{4}$$

An equation of the tangent to the circle at (3, 4) is therefore

$$y - 4 = -\frac{3}{4}(x - 3) \quad \text{or} \quad 3x + 4y = 25$$

Implicit Differentiation

(b) Solution 2 Solving the equation $x^2 + y^2 = 25$ for y, we get $y = \pm\sqrt{25 - x^2}$. The point (3, 4) lies on the upper semicircle $f(x) = \sqrt{25 - x^2}$ and so we consider the function $f(x) = \sqrt{25 - x^2}$ Differentiating f using the Chain Rule, we have

explicit differentiation

$$
\begin{aligned}
f'(x) &= \frac{1}{2}(25 - x^2)^{-\frac{1}{2}} \frac{d}{dx}(25 - x^2) \\
&= \frac{1}{2}(25 - x^2)^{-\frac{1}{2}}(-2x) \\
&= -\frac{x}{\sqrt{25 - x^2}}
\end{aligned}
$$

So

$$f'(3) = -\frac{3}{\sqrt{25 - 3^2}} = -\frac{3}{4}$$ → same result!

as in Solution 1, an equation of the tangent is $3x + 4y = 25$.

Example

(a) Find y' if $x^3 + y^3 = 6xy$.

(b) Find the tangent to $x^3 + y^3 = 6xy$ at the point (3, 3)

Solution

(a) Differentiate both sides of the equation with respect to x

$$\frac{d}{dx}(x^3 + y^3) = \frac{d}{dx}(6xy)$$

$$\frac{d}{dx}(x^3) + \frac{d}{dx}(y^3) = \frac{d}{dx}(6xy)$$

$$3x^2 + 3y^2y' = 6y + 6xy'$$

We now solve for y'

$$(y^2 - 2x)y' = 2y - x^2$$

$y'' = \frac{d}{dx}y'$

$$y' = \frac{2y - x^2}{y^2 - 2x}$$

Implicit Differentiation

(b) At point (3, 3), where $x = y = 3$

$$y' = \frac{2 \cdot 3 - 3^2}{3^2 - 2 \cdot 3} = -1$$

the tangent line is

$$y - 3 = -1(x - 3) \quad \text{or} \quad x + y = 6$$

Implicit Differentiation

Example

Find y' if $\sin(x+y) = y^2 \cos x$.

Solution

Differentiate both sides of the equation with respect to x

$$\cos(x+y)(1+y') = (2yy')(\cos x) + y^2(-\sin x)$$

$$\cos(x+y) + y^2 \sin x = (2y\cos x - \cos(x+y))y'$$

We now solve for y'

$$y' = \frac{\cos(x+y) + y^2 \sin x}{2y\cos x - \cos(x+y)}$$

contains both x and y

Example

Show that the tangent line to the ellipse ($a > 0$ and $b > 0$)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

point is on ellipse

at the point (x_0, y_0) is $\therefore \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} = 1$$

Solution

Differentiate both sides of the equation with respect to x

$$\frac{2x}{a^2} + \frac{2yy'}{b^2} = 0$$

We now solve for y'

$$y' = -\frac{b^2 x}{a^2 y}$$

The tangent line at (x_0, y_0) is

$$y - y_0 = y'(x_0, y_0)(x - x_0)$$

$$y - y_0 = -\frac{b^2 x_0}{a^2 y_0}(x - x_0)$$

$$\frac{yy_0 - y_0^2}{b^2} = -\frac{xx_0 - x_0^2}{a^2}$$

$$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2}$$

Note that (x_0, y_0) is a point on the ellipse, so that

$$\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$$

Therefore

$$\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$$

implies derivative

If $s = f(t)$ is the position function of a particle that is moving in a straight line, then

Velocity is defined as $v(t) = s' = \frac{ds}{dt}$. (More specifically, the rate of change of position with respect to time).
Acceleration is defined as $a(t) = v'(t) = s''(t)$. It represents the change of velocity with respect to time.
Speed is defined as the absolute value of velocity $|v(t)|$.

can never be negative

displacement → can be +ve or negative
distance → must be positive

The position of a particle is given by the equation

$$s = f(t) = t^3 - 6t^2 + 9t$$

where t is measured in seconds and s in meters. take note of units!

v m/s

a m/s^2

(a) Find the velocity at time t.

(b) When is the particle at rest?

(c) When is the particle moving forward (that is, in the positive direction)?

(d) Draw a diagram to represent the motion of the particle.

(e) Find the total distance traveled by the particle during the first 5 sec.

(f) Find the acceleration at time t and at 4 sec.

(g) Graph the position, velocity, and acceleration functions.

(h) When is the particle speeding up? When is it slowing down?

Rates of Change

(a) The velocity function is the derivative of the position function.

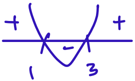
$$s = f(t) = t^3 - 6t^2 + 9t$$

$$v(t) = \frac{ds}{dt} = 3t^2 - 12t + 9$$

(b) The particle is at rest when $v(t) = 0$, that is

$$3t^2 - 12t + 9 = 0$$

$$3(t-1)(t-3) = 0$$



and this is true when $t = 1$ sec or $t = 3$ sec.
(c) The particle moves in the positive direction when $v(t) > 0$, that is,

$$3t^2 - 12t + 9 = 3(t-1)(t-3) > 0$$

This inequality is true when both factors are positive ($t > 3sec$) or when both factors are negative ($t < 1sec$).

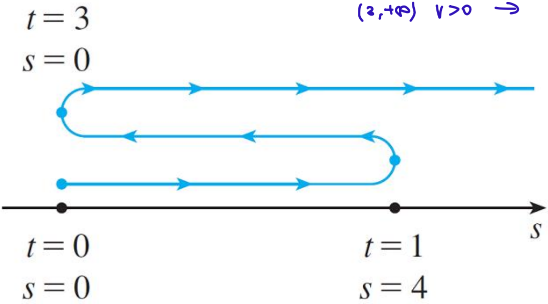
(d) For the above analysis, we see that the particle moves in the positive direction in the time intervals $t < 1$ and $t > 3$. It moves backward (in the negative direction) when $1 < t < 3$.
Since $s = f(t) = t^3 - 6t^2 + 9t$, we have $s(t = 0) = 0m$, $s(t = 1) = 4m$ and $s(t = 3) = 0m$.

direction of movement

$(0,1)$ $v>0$ $\rightarrow$

$(1,3)$ $v<0$ $\leftarrow$

$(3,+\infty)$ $v>0$ $\rightarrow$



(e) Find the total distance traveled by the particle during the first 5 sec? We need to calculate the distances traveled during the time intervals [0, 1], [1, 3], and [3, 5] separately.

*it is wrong to do $|f(5)-f(0)|$

The distance traveled in the first second is

$$|f(1) - f(0)| = |4 - 0| = 4m$$

From $t = 1$ to $t = 3$ the distance traveled is

$$|f(3) - f(1)| = |0 - 4| = 4m$$

From $t = 3$ to $t = 5$ the distance traveled is

$$|f(5) - f(3)| = |20 - 0| = 20m$$

The total distance is $4 + 4 + 20 = 28m$.

& always positive

(f) Find the acceleration at time t and at 4 sec?

The acceleration is the derivative of the velocity function
$v(t) = \frac{ds}{dt} = 3t^2 - 12t + 9$

$$a(t) = v' = \frac{dv}{dt} = 6t - 12$$

$$a(t = 4) = 6 \cdot 4 - 12 = 12\underline{m/sec^2}$$

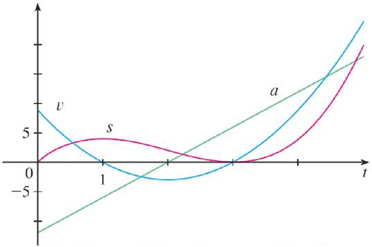
take note of units!

(g) Graph the position, velocity, and acceleration functions ?

$s = f(t) = t^3 - 6t^2 + 9t = t(t^2 - 6t + 9) = t(t-3)^2$

$v(t) = \frac{ds}{dt} = 3t^2 - 12t + 9$

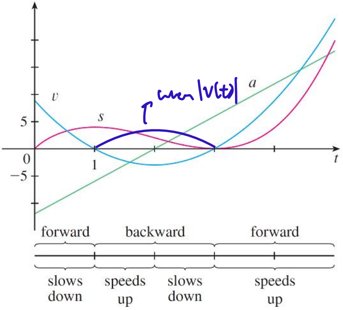
$a(t) = v' = \frac{dv}{dt} = 6t - 12$



(h) When is the particle speeding up? When is it slowing down?

Rates of Change

Speed is the absolute value of velocity $|v(t)|$



※ pay attention to units!

(h) The particle speeds up when the velocity is positive and increasing ($v(t)$ and $a(t)$ are both positive) or velocity is negative and decreasing ($v(t)$ and $a(t)$ are both negative).
In other words, the particle speeds up when the velocity and acceleration have the same sign. It happens when $1 < t < 2sec$ and when $t > 3sec$.

Related Rates

If we pump air into a balloon, both the volume and the radius of the balloon increase, and their rates of increase are highly related to each other.
But it is much easier to measure the increasing rate of the volume than the increasing rate of the radius.

The essential idea of related rate problem is to compute the rate of change of one quantity, in terms of the rate of change of another quantity (which may be more easily measured).

The procedure is to find an equation of the two quantities, and use the Chain Rule (to differentiate both sides with respect to time) to get the relation between their rates.

Example

$\frac{dV}{dt} = 100\ cm^3/s$

Air is being pumped into a spherical balloon so that its volume increases at a rate of $100cm^3/s$. How fast is the radius of the balloon increasing when the diameter is $50cm$?

want to find $\frac{dr}{dt}$ when $r = 25$

$d = 2r$

Solution We start by identifying two things:
1) The given information: the rate of increase of the volume of air is $100cm^3/s$;
2) The unknown: the rate of increase of the radius when the diameter is $50cm$.
In order to express these quantities mathematically, we introduce some suggestive notation:

Let V be the volume of the balloon and let r be its radius.

Related Rates

The key thing to remember is that rates of change are derivatives, and the volume and the radius are both functions of the time t in this problem.
The rate of increase of the volume with respect to time is the derivative $\frac{dV}{dt}$, and the rate of increase of the radius is $\frac{dr}{dt}$.
We can therefore restate the given and the unknown as follows:

$$\text{Given:} \quad \frac{dV}{dt} = 100cm^3/sec$$

$$\text{Unknown:} \quad \frac{dr}{dt} \quad \text{when} \quad r = 25cm$$

In order to connect $\frac{dV}{dt}$ and $\frac{dr}{dt}$, we first relate V and r by the formula for the volume of a sphere:

$$V = \frac{4}{3}\pi r^3$$

Related Rates

In order to use the given information, we differentiate each side of this equation with respect to t. To differentiate the right side, we need to use the Chain Rule:

$$\frac{dV}{dt} = \frac{dV}{dr}\frac{dr}{dt} = 4\pi r^2 \frac{dr}{dt}$$

Now we solve for the unknown quantity:

$$\frac{dr}{dt} = \frac{1}{4\pi r^2}\frac{dV}{dt} \to 100\,cm^3/s$$

If we put $r = 25$ and $dV/dt = 100$ in this equation, we obtain

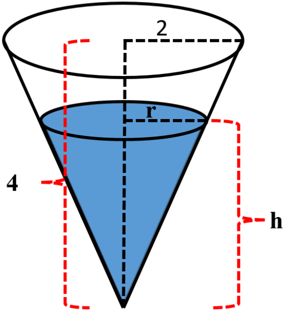
$$\frac{dr}{dt} = \frac{1}{4\pi(25)^2}100 = \frac{1}{25\pi}$$

The radius of the balloon is increasing at the rate of $\frac{1}{25\pi} \approx 0.0127 cm/s$.

Example

A water tank has the shape of an inverted circular cone with base radius 2 meter(m) and height 4 m. If water is being pumped into the tank at a rate of 2 m^3/min, find the rate at which the water level is rising when the water is 3 m deep.

Solution We first sketch the water tank in Figure below,



We let V, r and h be the volume, the surface radius, and the height of the water at time t (min). From the cone geometry, we can have

$$V = \frac{1}{3}\pi r^2 h$$

From the conditions, we can have $dV/dt = 2m^3/min$. The unknown is dh/dt at $h = 3m$.
In order to eliminate r, we use the similar triangles in the Figure,

$$\frac{r}{h} = \frac{2}{4} \Rightarrow r = \frac{h}{2}$$

therefore we have the equation for the water volume,

$$V = \frac{1}{3}\pi(\frac{h}{2})^2 h = \frac{\pi}{12}h^3$$

Related Rates

Now we have related the water volume V to the height of the water h, we can differentiate each side with respect to t,

$$\frac{dV}{dt} = \frac{\pi}{4}h^2\frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{4}{\pi h^2}\frac{dV}{dt}$$

Substituting $h = 3m$ and $dV/dt = 2m^3/min$ in to the above equation, we have

$$\frac{dh}{dt} = \frac{4}{\pi 3^2}2 = \frac{8}{9\pi}$$

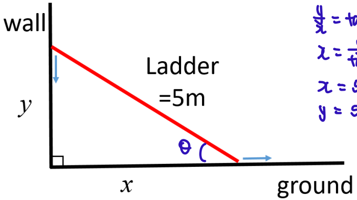
The water level is rising at a rate of $\frac{8}{9\pi} \approx 0.28m/min$.

Example

$\frac{dx}{dt} = 1$

A 5 meter(m) long ladder rests against a vertical wall. If the bottom of the ladder slides away from the wall at a rate of $1m/s$, how fast is the top of the ladder sliding down the wall when the bottom of the ladder is $3m$ from the wall?

Solution We first draw a diagram and label it in Figure below,



$\frac{y}{x} = \tan\theta$

$x = \frac{y}{\tan\theta}$

$x = 5 \cdot \cos\theta$

$y = 5 \cdot \sin\theta$

→ more complicated way to solve

Related Rates

We define x, y as the distance from the one tip of the ladder to the wall, and another tip of the ladder to the ground as in the Figure. Since the ladder is 5 m long, from the Pythagorean Theorem, we have

$$x^2 + y^2 = 5^2$$

Note that both x and y are function of t. Now we have $dx/dt = 1m/s$ and we need to know dy/dt at $x = 3m$.
Differentiating each side with respect to t using the chain rule, we have

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0 \Rightarrow \frac{dy}{dt} = -\frac{x}{y}\frac{dx}{dt}$$

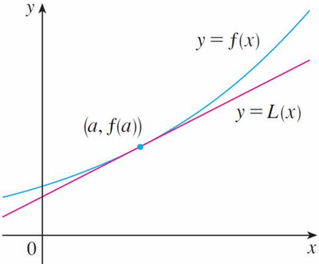
From the Pythagorean Theorem, when $x = 3m$, we have $y = 4m$ ($y = -4$ is unphysical). We can substitute these values and $dx/dt = 1m/s$ into the above equation, and have

$$\frac{dy}{dt} = -\frac{3}{4}(1) = -0.75m/s$$

Linear Approximations

Near the point of tangency, the curve always lies close to its tangent line. In fact, by zooming in toward a point on the graph of a differentiable function, we notice that the graph looks more and more like its tangent line. This observation provides a way to approximate values of functions. The idea is that it might be easy to calculate the value of function $f(x)$ at $x = a$, but difficult (or even impossible) to compute $f(x)$ at nearby points $(x \approx a)$.

So we approximate the $f(x)$ at $(x \approx a)$ by the linear function $L(x)$, whose graph is the tangent line of $f(x)$ at $(a, f(a))$.



In other words, we use the tangent line at $(a, f(a))$ as an approximation to the curve $y = f(x)$ when x is near a. An equation of this tangent line is

$y - f(a) = f'(a)(x-a)$

$$y = f(a) + f'(a)(x - a)$$

The linear approximation or tangent line approximation of f at a is

Linear Approximation or Tangent Line Approximation

$$f(x) \approx f(a) + f'(a)(x - a)$$ when x is near a

The linear function is the tangent line,

Linearization of Function f at a

$\downarrow$

$y = ax + b$

$$L(x) = f(a) + f'(a)(x - a)$$

Example

Find a linear approximation of function $f(x) = \sqrt{x+3}$ at $a = 1$ and use it to approximate the numbers $\sqrt{3.98}$ and $\sqrt{4.05}$. Are these approximations overestimates or underestimates? $L(x) = f(a) + f'(a)(x-a)$

Solution The derivative of $f(x) = \sqrt{x+3}$ is

$$f'(x) = \frac{1}{2}(x+3)^{-\frac{1}{2}} = \frac{1}{2\sqrt{x+3}}$$

so we have $f(1) = 2$ and $f'(1) = \frac{1}{4}$. The linear approximation of function $f(x)$ at $x = 1$ is

$$L(x) = f(1) + f'(1)(x-1) = 2 + \frac{1}{4}(x-1) = \frac{7}{4} + \frac{x}{4}$$

The corresponding linear approximation is

$$\sqrt{x+3} \approx \frac{7}{4} + \frac{x}{4}$$

x is near $a=1$

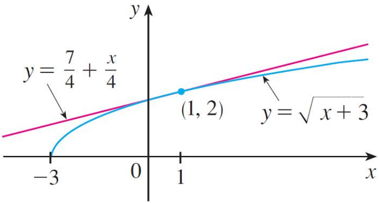
Linear Approximations

In particular, we have

$$\sqrt{3.98} \approx \frac{7}{4} + \frac{0.98}{4} = 1.995$$

and

$$\sqrt{4.05} \approx \frac{7}{4} + \frac{1.05}{4} = 2.0125$$



The two approximations are all overestimates, because the tangent line lies above the curve.

Notice that the tangent line approximation gives good estimates when x is close to 1, but the accuracy of the approximation deteriorates when x is farther away from 1.

a=1

	x	From $L(x)$	Actual value	error
$\sqrt{3.9}$	0.9	1.975	1.97484176...	≈ 0.001
$\sqrt{3.98}$	0.98	1.995	1.99499373...	≈ 0.00001
$\sqrt{4}$	1	2	2.00000000...	
$\sqrt{4.05}$	1.05	2.0125	2.01246117...	≈ 0.0001
$\sqrt{4.1}$	1.1	2.025	2.02484567...	≈ 0.001
$\sqrt{5}$	2	2.25	2.23606797...	≈ 0.02
$\sqrt{6}$	3	2.5	2.44948974...	≈ 0.1

very far from a=1

Linear Approximations

Example

Use a linear approximation to estimate the given number

(a) $\sqrt{99.8}$

(b) $\cos(29^{\circ})$ degree

Solution (a) We can let $f(x) = \sqrt{x}$. We know $\sqrt{100} = 10$ and $99.8 \approx 100$. Therefore, we can set $a = 100$ and use $L(x) = f(a) + f'(a)(x - a)$ to evaluate $f(99.8)$. Since $f'(x) = \frac{1}{2}x^{-1/2}$, we have,

$$f(99.8) \approx f(100) + f'(100)(99.8 - 100) = 10 - 1/20 * 0.2 = 9.99$$

Linear Approximations

30°

(b) We can let $f(x) = \cos(x)$. We know $29^o = \frac{29}{180}\pi \approx \frac{\pi}{6}$. We set $a = \frac{\pi}{6}$ and use $L(x) = f(a) + f'(a)(x - a)$ to evaluate $f(\cancel{99.8})$. Since $f'(x) = -\sin(x)$, we have,

29°

*have to do in radians

$$f(29^o) = f(\frac{29}{180}\pi) \approx f(\frac{\pi}{6}) + f'(\frac{\pi}{6})(\frac{29}{180}\pi - \frac{\pi}{6})$$

$$f(29^o) \approx \cos(\frac{\pi}{6}) - \sin(\frac{\pi}{6})(\frac{-\pi}{180})$$

$$f(29^o) \approx \frac{\sqrt{3}}{2} + \frac{\pi}{360}$$

Differentials

The ideas behind linear approximations can be better understood through differentials.
For a differentiable function $y = f(x)$, we have $y' = f'(x) = \frac{dy}{dx}$. We can treat the differential dx as an independent variable (i.e., treat dx as a variable, such as x, y, z, t, etc), the differential dy is then a function of dx, and is defined in terms of dx by the equation

$$dy = f'(x)dx$$

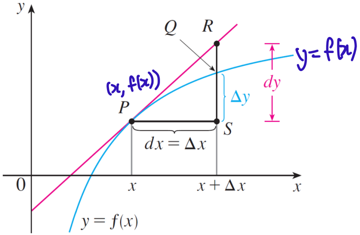
Differentials

For a point $P(x, f(x))$ on the graph of $f(x)$, we increase the coordinate x by a small value Δx, and arrive at a new point $Q(x + \Delta x, f(x + \Delta x))$ on the graph. We let $dx = \Delta x$, and have

The change of y **is** $\Delta y = f(x + \Delta x) - f(x)$.

The differential dy **is** $dy = f'(x)dx$.

Geometrically, dy represents the amount that the tangent line rises or falls (the change in the linearization), when x changes by an amount dx. Whereas Δy represents the amount that the curve $y = f(x)$ rises or falls, when x changes by an amount $dx = \Delta x$.



Differentials

Example

Compare the values of Δy and dy if $y = f(x) = x^3 + x^2 - 2x + 1$ and x changes (a) from 2 to 2.05 and (b) from 2 to 2.01.

Solution

(a) We have

$$f(2) = 2^3 + 2^2 - 2(2) + 1 = 9$$

$$f(2.05) = (2.05)^3 + (2.05)^2 - 2(2.05) + 1 = 9.717625$$

$$\Delta y = f(2.05) - f(2.0) = 0.717625$$

In general,

$$dy = f'(x)dx = (3x^2 + 2x - 2)dx$$

When $x = 2$ and $dx = \Delta x = 0.05$, this becomes

$$dy = [3(2)^2 + 2(2) - 2]0.05 = 0.7$$

error ≈ 0.02

Differentials

(b) We have

$$f(2.01) = (2.01)^3 + (2.01)^2 - 2(2.01) + 1 = 9.140701$$

$$\Delta y = f(2.01) - f(2.0) = 0.140701$$

When $x = 2$ and $dx = \Delta x = 0.01$, this becomes

error ≈ 0.001

$$dy = [3(2)^2 + 2(2) - 2]0.01 = 0.14$$

From the example, we can find that the differential dy can be used as an approximation of Δy.

Differentials

Example

The radius of a sphere was measured and found to be $21cm$ with a possible error in measurement of at most $0.05cm$. What is the maximum error in using this value of the radius to compute the volume of the sphere?

Solution If the radius of the sphere is r, then its volume is $V = \frac{4}{3}\pi r^3$. If the error of r (in the measured value) is denoted by $dr = \Delta r$, then the corresponding error in the calculated value of V is ΔV, which can be approximated by the differential

$\Delta V \approx dV = f'(v)\,dr$

$$dV = V'dr = 4\pi r^2 dr$$

When $r = 21$ and $dr = 0.05$, this becomes

$$dV = 4\pi(21)^2 \cdot 0.05 \approx 277$$

The maximum error in the calculated volume is about $277cm^3$.

Note

Although the possible error in the above example may appear to be rather large, a better picture of the error is given by the relative error, which is computed by dividing the error by the total volume

$$\frac{\Delta V}{V} \approx \frac{dV}{V} = \frac{4\pi r^2 dr}{\frac{4}{3}\pi r^3} = 3\frac{dr}{r}$$

Thus the relative error in the volume is about three times the relative error in the radius. The relative error in the radius is approximately $\frac{dr}{r} = 0.05/21 \approx 0.0024$. The relative error of the volume is about $3\frac{dr}{r} = 0.007$.

The errors could also be expressed as percentage errors of 0.24% in the radius and 0.7% in the volume.